SOFTWARE ENGINEERING

ASSIGNMENT

SMART PET CARE AND VETERINARY\nMANAGEMENT PLATFORM

A Software Engineering System Design and Analysis

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Academic Assignment

1. Abstract

The Smart Pet Care and Veterinary Management Platform is a proposed digital system that helps pet owners and veterinary clinics manage pet health and routine care in one place. It stores pet profiles, vaccination histories, medical records, prescriptions, appointments, medication schedules, grooming activities, and other preventive-care tasks. The platform provides reminders for upcoming appointments, vaccinations, medicines, and routine activities. Authorized veterinary professionals can review and update relevant medical information, while pet owners can access their pets' records and communicate with the clinic. The system aims to reduce missed care activities, improve record organization, support timely preventive care, and make communication more efficient. This assignment analyses the problem, identifies system requirements, proposes a modular architecture and database design, and describes implementation, security, testing, and expected outcomes.

2. Introduction

Pet care involves several recurring responsibilities, including vaccinations, health check-ups, medication, grooming, nutrition-related notes, and emergency visits. When this information is written in notebooks, stored in different applications, or remembered manually, important dates and records can be missed. Veterinary clinics also need a dependable way to maintain patient histories, manage appointment schedules, and share updates with pet owners.

The proposed platform brings these activities together in a single application. Each pet has a digital profile linked to its owner. Health records and care schedules can be maintained over time, and a reminder service alerts owners about upcoming tasks. A veterinarian or authorized clinic staff member can update records according to their assigned permissions. The platform is intended to support—not replace—professional veterinary advice.

3. Industry Context

Pet owners increasingly expect convenient digital access to appointments and health information. Veterinary clinics need accurate, searchable records and a consistent communication process. A centralized platform can reduce repeated data entry, make historical information easier to retrieve, and help owners follow preventive-care plans. It can also produce summaries that support consultations and routine clinic administration.

- Pet owners need a clear view of each pet's profile, vaccinations, prescriptions, and upcoming care activities.
- Veterinary professionals need reliable access to relevant medical history and a controlled way to update records.
- Clinics need appointment management, status tracking, and a way to send useful updates to owners.
- Both groups benefit from timely reminders, organized records, and reports that can be reviewed when needed.

4. Problem Statement

Design and develop a Smart Pet Care and Veterinary Management Platform that maintains pet profiles, vaccination histories, veterinary records, appointments, medication schedules, and routine care activities. The system should provide reminders for upcoming veterinary appointments and scheduled preventive-care activities while allowing authorized veterinary professionals to update relevant records.

The core problem is the lack of a unified and dependable process for tracking a pet's care lifecycle. The proposed solution must keep information organized, help users act on time, and protect private owner and medical data.

5. Objectives

- Maintain pet profiles containing identification and basic health information.
- Record vaccination details, dates, and recommended next doses.
- Manage veterinary appointments, including booking, rescheduling, cancellation, and visit status.
- Maintain medical records such as diagnoses, consultation notes, prescriptions, and test summaries.
- Generate reminders for appointments, vaccinations, medication, grooming, and other preventive-care tasks.
- Track medication schedules, dosage instructions entered by a veterinary professional, and completion status.
- Manage grooming schedules and routine-care activities.
- Enable communication between pet owners and veterinary clinics.
- Generate useful pet-care reports and summaries.
- Apply role-based access so only authorized users can view or update restricted information.

6. Scope of the System

6.1 Included in Scope

- Owner registration, login, and profile management.
- Creation and maintenance of multiple pet profiles under one owner.
- Vaccination and medical-history recording.
- Appointment booking and clinic-side appointment management.
- Medication, grooming, and preventive-care schedules.
- Reminder notifications and a dashboard of upcoming tasks.
- Secure communication and record updates by authorized veterinary staff.
- Basic reports for an individual pet or a selected date range.

6.2 Outside the Initial Scope

- Automatic diagnosis of diseases or replacement of a veterinarian's clinical judgement.
- Unsupervised changes to prescriptions or medication dosage.
- Emergency response services or guaranteed real-time medical monitoring.
- Integration with every external clinic, laboratory, pharmacy, or wearable device from the first release.

7. Stakeholders and User Roles

Stakeholder / Role	Main Responsibilities	Access Level
Pet Owner	Create pet profiles, view records, book appointments, track care tasks, receive reminders, contact the clinic.	Own pets and related appointments/records
Veterinarian	Review history, add consultation notes, update vaccination and treatment records, provide follow-up instructions.	Authorized clinical records
Clinic Administrator / Receptionist	Manage clinic details, appointment slots, booking status, and owner communication.	Scheduling and administrative data
System Administrator	Manage accounts, roles, system settings, backups, and audit	Administrative access subject

	logs.	to policy

8. Functional Requirements

ID / Requirement	Description
FR-01: User account management	The system shall allow owners and clinic staff to register or sign in securely and maintain their profile information.
FR-02: Pet profile management	The system shall allow an owner to add, view, edit, and archive pet profiles. Each profile shall have a unique identifier.
FR-03: Vaccination tracking	The system shall store vaccine name, administration date, administering clinic, notes, and next-due date when provided.
FR-04: Appointment management	The system shall display available slots and support booking, rescheduling, cancellation, and status updates.
FR-05: Medical records	Authorized veterinary staff shall be able to create and update consultation notes, diagnoses, treatment details, and attached documents.
FR-06: Medication schedule	The system shall record medicine name, prescribed instructions, start/end dates, and reminder times as entered by an authorized professional.
FR-07: Routine-care schedule	The system shall track grooming and other recurring care activities, including due dates and completion status.
FR-08: Notifications	The system shall generate reminders based on configured dates and times and record whether a reminder was sent.
FR-09: Communication	The system shall provide a secure channel for owners to send messages or questions to the clinic and receive responses.
FR-10: Reports	The system shall generate summaries of vaccinations, appointments, medical visits, medication schedules, and upcoming care tasks.
FR-11: Access control	The system shall restrict actions according to role and ownership and record important changes to clinical records.
FR-12: Search and retrieval	The system shall allow authorized users to search records using pet name, owner, record type, or date range.

9. Non-Functional Requirements

Quality Attribute	Requirement
Security	Use secure authentication, role-based authorization, encrypted connections, and protected storage for sensitive records.
Privacy	Show owners only their own pets' information; expose clinical information to authorized clinic personnel only.
Reliability	Avoid losing saved records and maintain consistent appointment and reminder data.
Performance	Common screens and searches should respond within a reasonable time under expected student-project workloads.
Usability	Use clear navigation, readable labels, simple forms, and mobile-friendly pages for owners and clinic staff.
Maintainability	Separate the system into modules so features can be tested and changed independently.
Scalability	Design the database and services to support more owners, pets, clinics, and records over time.
Availability and recovery	Use backups and a documented recovery process to reduce the impact of service or data failures.
Auditability	Record who changed important clinical data and when the change occurred.

10. Proposed System Architecture

A three-layer architecture is recommended because it separates the user interface, business rules, and data storage. This separation makes the system easier to test, maintain, and extend.

Layer	Components	Purpose
Presentation Layer	Owner dashboard, veterinarian dashboard, appointment screens, care calendar, reports.	Collects input and displays records and reminders.
Application / Service Layer	Authentication, pet service, appointment service, medical-record service, reminder service, reporting service.	Validates requests, applies business rules, checks permissions, and coordinates workflows.
Data Layer	Relational database, document/file storage, audit log, backup process.	Stores structured information and supports safe retrieval and recovery.
Notification Integration	Email, SMS, or in-app notification provider.	Delivers reminders and communication alerts using configured channels.

Example technology stack: HTML, CSS, and JavaScript for a web interface; Python with Flask or Django for the backend; and MySQL or PostgreSQL for the database. A student prototype may use SQLite initially. Email or in-app notifications can be implemented first, with SMS integration added later. The technology choice can be adjusted to match the course environment.

11. Major System Modules

11.1 User and Access Management

Handles registration, login, profile details, password protection, role assignment, and session management.

11.2 Pet Profile Management

Stores pet name, species, breed, date of birth or estimated age, sex, weight history, identification details, allergies, and owner contact information where appropriate.

11.3 Vaccination Management

Stores vaccine history, administered date, clinic/veterinarian, notes, and next due date. It identifies upcoming and overdue vaccinations based on recorded dates.

11.4 Appointment Management

Shows available clinic slots, checks for conflicting bookings, stores appointment purpose, and supports confirmation, rescheduling, cancellation, and visit completion.

11.5 Medical Record Management

Stores consultation date, symptoms reported, clinical findings, diagnosis, treatment plan, prescriptions, test results, and follow-up notes. Updates are restricted to authorized professionals.

11.6 Medication Tracking

Displays medication instructions and reminder times that were entered by the veterinarian. Owners can mark a scheduled reminder as completed or missed; the application must not independently alter a prescribed dosage.

11.7 Grooming and Preventive Care

Tracks bathing, nail trimming, dental care, parasite prevention, and other owner- or veterinarian-recommended routine activities.

11.8 Reminder and Notification Service

Checks due dates, creates reminders, prevents duplicate alerts where possible, and stores delivery status. Users can configure suitable notification preferences.

11.9 Communication Module

Provides a secure message thread between an owner and a clinic. It can support appointment questions and follow-up clarifications, while warning that emergencies require direct veterinary or emergency services.

11.10 Reporting Module

Creates pet health summaries, vaccination history reports, appointment history, medication schedule views, and lists of upcoming or overdue care tasks.

12. Database Design

A relational database is suitable because owners, pets, appointments, and medical records have clear relationships. Each table should use a primary key, and foreign keys should connect related records. The following is a proposed high-level schema; field types can be selected during implementation.

Table	Important Fields	Purpose / Relationship
Users	user_id (PK), full_name, email, phone, password_hash, role, created_at	Stores owner and clinic-user accounts. Passwords are stored as secure hashes, not plain text.
Pets	pet_id (PK), owner_id (FK), pet_name, species, breed, date_of_birth, sex, notes	Each pet belongs to an owner; one owner can have multiple pets.
Vaccinations	vaccination_id (PK), pet_id (FK), vaccine_name, administered_on, next_due_on, clinic_id, notes	Stores a pet's vaccination history and due dates.
Clinics	clinic_id (PK), clinic_name, address, phone, email	Stores veterinary clinic details.
Appointments	appointment_id (PK), pet_id (FK), owner_id (FK), clinic_id (FK), vet_id (FK), appointment_at, reason, status	Connects a pet and owner to a clinic and optional assigned veterinarian.
MedicalRecords	record_id (PK), pet_id (FK), appointment_id (FK), vet_id (FK), record_date, diagnosis, treatment_notes, follow_up_date	Stores consultation and treatment records.
Medications	medication_id (PK), pet_id (FK), record_id (FK), medicine_name, instructions, start_date, end_date, reminder_times	Stores veterinarian-entered medication instructions and reminder schedule.
CareTasks	task_id (PK), pet_id (FK), task_type, description, due_at, repeat_rule, status	Tracks grooming and preventive-care tasks.
Notifications	notification_id (PK), user_id (FK), task_id (FK, nullable), appointment_id (FK, nullable), scheduled_at, sent_at, status	Tracks reminder scheduling and delivery state.
Messages	message_id (PK), sender_id (FK), recipient_id (FK), clinic_id (FK), message_text, sent_at, read_at	Stores communication messages with access controls.
AuditLogs	audit_id (PK), user_id (FK), action, entity_type, entity_id, timestamp	Tracks sensitive record changes for accountability.

13. Key Relationships and Business Rules

- One owner can register many pets, but every pet profile must be linked to an owner account.
- One pet can have many vaccinations, appointments, medical records, medications, and care tasks.
- An appointment must be associated with a pet, owner, and clinic; an assigned veterinarian may be optional until the clinic confirms the booking.
- The system should prevent overlapping bookings for the same veterinarian or appointment slot.
- Only authorized veterinary staff may create or modify clinical findings and prescription instructions.
- A reminder should be generated from a stored due date or schedule and should have a status such as pending, sent, failed, completed, or cancelled.
- Deleting a user or pet should not silently remove legally or operationally important medical history; archival and retention rules should be defined.
- All dates and times should be stored consistently, with the user interface showing the appropriate local time.

14. Typical System Workflow

1. The pet owner registers, signs in, and creates a pet profile.
2. The owner adds available vaccination history or imports records entered by a clinic.
3. The owner views clinic availability and requests an appointment for the selected pet.
4. The clinic confirms the appointment and the system displays it on both relevant dashboards.
5. After the consultation, the authorized veterinarian updates the medical record and adds any follow-up date or medication instructions.
6. The reminder service checks upcoming appointment, vaccination, medication, grooming, and preventive-care dates.
7. The owner receives reminders and can view the task, mark a routine task as completed, or contact the clinic when clarification is needed.
8. The owner or veterinarian generates a report when a health-history summary is required.

15. Use Cases

Use Case	Primary Actor	Precondition	Main Result
Create Pet Profile	Pet Owner	Owner is signed in.	A pet profile is saved and linked to the owner.
Book Appointment	Pet Owner	Pet profile exists and a slot is available.	Appointment request is created and confirmation status is shown.
Update Medical Record	Veterinarian	Veterinarian is authenticated and authorized for the record.	A dated clinical entry is saved and audit information is recorded.
Record Vaccination	Veterinarian / Authorized Clinic Staff	Correct pet is selected.	Vaccination history is updated and next due date can be scheduled.
Receive Reminder	Pet Owner	A future task or appointment has a due date.	A notification is created and delivery status is tracked.
Generate Care Report	Pet Owner / Veterinarian	User has permission to view the pet's data.	A summary of selected records is displayed or exported.
Send Clinic Message	Pet Owner / Clinic Staff	User is authenticated and recipient is	A message is stored and

		valid.	the recipient can be notified.

16. Reminder Logic

The reminder service should periodically compare scheduled dates with the current time. For a prototype, this can be implemented with a scheduled backend job. In a production system, a reliable job queue or scheduler is preferred.

9. Read appointments and care tasks whose reminder time has arrived and whose status is still pending.
10. Check that the related owner account is active and has a valid notification preference.
11. Create a notification record and send the alert through the selected channel.
12. Update the notification status to sent or failed and store a timestamp.
13. Avoid repeated alerts by using a unique task/reminder identifier and tracking whether the alert has already been sent.
14. Allow users to review upcoming, completed, missed, and overdue activities in the dashboard.

Example: if a vaccination record has a next-due date of 20 November, the system may schedule a configurable advance reminder and a due-date reminder. The exact interval should be configurable and should follow the clinic's guidance rather than assuming that every vaccine follows the same schedule.

17. User Interface Design

- Owner Dashboard: pet cards, next appointment, upcoming vaccinations, medication reminders, and routine-care tasks.
- Pet Profile Page: identification details, health notes, vaccination timeline, medical-record history, and attachments.
- Appointment Calendar: available slots, booking status, appointment reason, and reschedule/cancel actions.
- Veterinarian Dashboard: daily appointments, patient history, pending record updates, and follow-up tasks.
- Reminder Centre: due soon, overdue, completed, and failed notifications.
- Reports Page: date filters, pet selection, and downloadable health summaries.

The interface should use plain language, clear status labels, accessible colour contrast, and responsive layouts for mobile phones and desktop computers. Important actions such as cancelling an appointment should request confirmation.

18. Security, Privacy, and Ethical Considerations

- Use a secure password-hashing algorithm and never store plain-text passwords.
- Use role-based access control and verify permissions on the server for every protected request.
- Use HTTPS in deployed environments and protect backups and uploaded documents.
- Collect only information required for pet care and clinic operations, and explain how it is used.
- Record access and changes to important medical records in an audit log.
- Validate user input, restrict file types and sizes, and defend against common web vulnerabilities.
- Provide a process to correct inaccurate records while preserving appropriate change history.
- Do not present automated reminders or generated summaries as a medical diagnosis.
- Make clear that the platform is not an emergency service and that urgent health concerns require direct professional help.

19. Testing Strategy

Testing should confirm that individual modules work correctly and that the complete workflow behaves as expected. Testing should include normal inputs, invalid inputs, access-control checks, and boundary cases.

Test ID	Test Scenario	Expected Result
TC-01	Register with valid required details.	Account is created and the user can sign in.
TC-02	Attempt to open another owner's pet record.	Access is denied and the event may be logged.
TC-03	Create a pet with required fields missing.	Validation message appears and invalid data is not saved.
TC-04	Book an available appointment slot.	Booking is stored and status is displayed correctly.
TC-05	Book a slot already reserved for the same veterinarian.	System rejects the conflicting booking or requests another slot.
TC-06	Veterinarian adds a vaccination record.	Record is saved under the correct pet and next-due information is available.
TC-07	Create a task with a reminder due soon.	A reminder is generated at the configured time and status is tracked.
TC-08	Unauthorized owner attempts to edit a prescription.	Edit is blocked.
TC-09	Generate a pet-care report for a date range.	Report includes only records in the selected range and within the user's permissions.
TC-10	Notification provider fails to deliver an alert.	Failure is recorded and retry or alternative handling follows the configured policy.

20. Implementation Plan

15. Requirement Analysis: confirm users, core features, data fields, reminder rules, and privacy needs.
16. System Design: prepare use cases, database relationships, screen wireframes, and service/module boundaries.
17. Database Setup: create tables, keys, relationships, validation constraints, and sample data.
18. Core Development: implement authentication, pet profiles, vaccination records, and appointment management.
19. Care and Reminder Development: implement medication schedules, routine-care tasks, and notification status tracking.
20. Clinic Features: add veterinarian access, medical-record updates, and owner-clinic communication.
21. Reporting: create pet-care summaries and upcoming/overdue task reports.
22. Testing and Review: execute test cases, correct defects, review access permissions, and collect user feedback.
23. Deployment and Maintenance: deploy to a suitable server, configure backups, monitor errors, and update the application as needed.

21. Risks and Mitigation

Risk	Possible Impact	Mitigation
Incorrect or incomplete data	Care reminders or reports may be misleading.	Validate fields, identify missing information, and let authorized users correct records.
Missed notification	Owner may overlook an appointment or care task.	Track delivery status, allow dashboard review, and provide retry or alternative notification options.
Unauthorized access	Private owner or medical information may be exposed.	Use server-side authorization, secure authentication, audit logs, and security testing.
Duplicate appointments	Scheduling confusion for owners and clinic staff.	Check slot availability transactionally before confirming a booking.

System or database failure	Temporary loss of access or risk to records.	Maintain backups, test recovery, and monitor service health.
Over-reliance on reminders	Users may treat the app as a substitute for professional advice.	Display clear limitations and direct users to veterinarians for clinical decisions.

22. Expected Outcomes and Benefits

- Improved pet-care management through one organized record system.
- Reduced likelihood of missed vaccination schedules through date-based reminders.
- Better organization of veterinary records and easier retrieval during consultations.
- Improved communication between pet owners and veterinary clinics.
- Timely preventive care through appointment, medication, grooming, and routine-care tracking.
- Improved pet-owner experience through a clear dashboard and access to relevant records.
- More efficient clinic administration through organized appointment schedules and searchable history.
- Better visibility of upcoming and overdue tasks through reports and status indicators.

23. Future Enhancements

- Mobile application for Android and iOS with secure sign-in.
- Calendar integration and configurable email, SMS, or push notifications.
- Online appointment confirmation and clinic-specific availability management.
- Secure upload of laboratory reports, prescriptions, and scanned vaccination cards.
- Multiple-clinic support and controlled transfer of records when an owner changes clinic.
- Optional wearable-device integration for activity tracking, subject to device compatibility and owner consent.
- Multilingual interface and accessibility improvements.
- Analytics that summarize appointment patterns and overdue preventive-care tasks without attempting to diagnose disease.

24. Conclusion

The Smart Pet Care and Veterinary Management Platform provides a structured approach to managing pet profiles, vaccination histories, appointments, medical records, medication schedules, grooming, preventive-care activities, and owner-clinic communication. A modular architecture, relational database, role-based access control, and reliable reminder workflow can make the platform practical to implement and maintain. Its main value is to help owners and veterinary clinics keep information organized and act on scheduled care in a timely manner. The system should be designed as a support tool, with clinical decisions and prescription instructions remaining under the responsibility of qualified veterinary professionals.

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